SQL - PIZZA HUT SALES ANALYSIS

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PROBLEM STATEMENT

Pizza Hut seeks to obtain valuable insights into its overall sales performance, customer preferences, and revenue trends. With a high volume of transactions and a wide variety of pizza categories, identifying the best-selling products, key revenue drivers, and evolving customer behavior poses a significant challenge. The objective of this project is to analyze historical sales data using SQL to uncover meaningful patterns and trends. These insights will help Pizza Hut make informed decisions related to menu optimization, inventory planning, and marketing strategies. Ultimately, this analysis aims to boost profitability, enhance customer satisfaction, and support data-driven business operations.

OBJECTIVE

The primary objective is to leverage SQL to conduct a comprehensive analysis of Pizza Hut's sales data, providing data-driven recommendations to boost revenue, enhance operational efficiency, and optimize menu performance.

- Identify the most popular pizzas and categories based on the total number of orders.
- Determine the top revenue-generating pizzas to highlight high-margin products.
- Analyze customer purchasing behavior, including peak ordering hours and preferred pizza sizes.
- Evaluate category-wise sales performance to enhance menu planning and inventory management.
- Provide data-driven recommendations aimed at increasing revenue, minimizing waste, and improving overall operational efficiency.

BUSINESS CONTEXT

It is focused on understanding customer behavior and purchasing trends to drive actionable recommendations. The goal is to strengthen Pizza Hut's understanding of customer preferences and purchasing trends through data analysis.

The analysis specifically aims to provide actionable insights that will:

- Menu & Revenue Optimization
- Prioritize High-Value Items: Identify both the most popular pizzas and the highest-margin, revenue-generating pizzas (e.g., premium Chicken options) to focus marketing efforts and increase profitability.
- Optimize Menu Mix: Recognize which pizzas and categories are most in demand to streamline the menu and enhance the customer experience.
- Operational Efficiency & Waste Reduction:
- Dynamic Inventory Management: Adjust inventory and ingredient levels based on demand forecasts for popular pizzas, ensuring stockouts are minimized and food waste is reduced.
- Effective Resource Planning: Plan staffing and delivery resources efficiently by accurately identifying peak ordering times (hours, days, seasons).
- Targeted Customer Engagement:
- Increase Sales Volume: Implement highly targeted promotions and offers that align with proven customer preferences, ultimately driving higher sales volume and improving customer satisfaction.

Data Overview and Schema

This project is built upon a fictional relational dataset designed for sales analysis. It consists of four distinct CSV files that collectively capture the transactional data, menu structure, and order specifics necessary for comprehensive analysis.

Source: The complete dataset is accessible via the following link: https://drive.google.com/drive/folders/14IaODHUAeZX548A74cDEunsmNVL6znlc?usp=drive

1.Pizzas:

pizza_id(str): The unique identifier for each specific size-and-type combination of a pizza (Primary Key).

pizza_type_id(Str): An identifier that links this specific pizza variant to the general pizza name and category details found in the **pizza_types table (**Foreign Key to pizza_types.csv)

Size (String/Char): The physical size of the pizza. Common values include S (Small), M (Medium), and L (Large).

Price (Decimal/Numeric): The price of the pizza, which is determined by its unique size and type combination.

pizza_id	pizza_type_id	size	price
bbq_ckn_s	bbq_ckn	S	12.75
bbq_ckn_m	bbq_ckn	M	16.75
bbq_ckn_l	bbq_ckn	L	20.75
cali_ckn_s	cali_ckn	S	12.75
cali_ckn_m	cali_ckn	M	16.75
cali_ckn_l	cali_ckn	L	20.75
ckn_alfredo_s	ckn_alfredo	S	12.75
ckn_alfredo_m	ckn_alfredo	M	16.75
ckn_alfredo_l	ckn_alfredo	L	20.75
ckn_pesto_s	ckn_pesto	S	12.75
ckn_pesto_m	ckn_pesto	M	16.75
ckn_pesto_l	ckn_pesto	L	20.75
southw_ckn_s	southw_ckn	S	12.75
southw_ckn_m	southw_ckn	M	16.75
southw_ckn_l	southw_ckn	L	20.75
thai_ckn_s	thai_ckn	S	12.75

2. Pizza_Types:

pizza_type_id(String): The unique identifier for the general pizza type, regardless of size. Primary Key, Foreign Key to pizzas.csv

Name (String): The official, market-facing name of the pizza type.

Category (String): The broad classification or group the pizza belongs to (e.g., Chicken, Veggie, Classic, Supreme).

Ingredients (String): A detailed, comma-separated list of all primary ingredients used to make this pizza type.

pizza_type_id	name	category	ingredients
bbq_ckn	The Barbecue Chicken Pizza	Chicken	Barbecued Chicken, Red Peppers, Green Peppe...
cali_ckn	The California Chicken Pizza	Chicken	Chicken, Artichoke, Spinach, Garlic, Jalapeno P...
ckn_alfredo	The Chicken Alfredo Pizza	Chicken	Chicken, Red Onions, Red Peppers, Mushrooms...
ckn_pesto	The Chicken Pesto Pizza	Chicken	Chicken, Tomatoes, Red Peppers, Spinach, Garl...
southw_ckn	The Southwest Chicken Pizza	Chicken	Chicken, Tomatoes, Red Peppers, Red Onions, ...
thai_ckn	The Thai Chicken Pizza	Chicken	Chicken, Pineapple, Tomatoes, Red Peppers, T...
big_meat	The Big Meat Pizza	Classic	Bacon, Pepperoni, Italian Sausage, Chorizo Sau...
classic_dlx	The Classic Deluxe Pizza	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppe...
hawaiian	The Hawaiian Pizza	Classic	Sliced Ham, Pineapple, Mozzarella Cheese
ital_cpcllo	The Italian Capocollo Pizza	Classic	Capocollo, Red Peppers, Tomatoes, Goat Chee...
napolitana	The Napolitana Pizza	Classic	Tomatoes, Anchovies, Green Olives, Red Onion...
pep_msh_pep	The Pepperoni, Mushroom, ...	Classic	Pepperoni, Mushrooms, Green Peppers
pepperoni	The Pepperoni Pizza	Classic	Mozzarella Cheese, Pepperoni
the_greek	The Greek Pizza	Classic	Kalamata Olives, Feta Cheese, Tomatoes, Garli...
brie_carre	The Brie Carre Pizza	Supreme	Brie Carre Cheese, Prosciutto, Caramelized Oni...
calabrese	The Calabrese Pizza	Supreme	'Nduja Salami, Pancetta, Tomatoes, Red Onions...
ital_supr	The Italian Supreme Pizza	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red O...
peppr_salami	The Pepper Salami Pizza	Supreme	Genoa Salami, Capocollo, Pepperoni, Tomatoes,...
prsc_argla	The Prosciutto and Arugula ...	Supreme	Prosciutto di San Daniele, Arugula, Mozzarella C...
sicilian	The Sicilian Pizza	Supreme	Coarse Sicilian Salami, Tomatoes, Green Olives, ...
soppressata	The Soppressata Pizza	Supreme	Soppressata Salami, Fontina Cheese, Mozzarell...
spicy_ital	The Spicy Italian Pizza	Supreme	Capocollo, Tomatoes, Goat Cheese, Artichokes,...
spinach_supr	The Spinach Supreme Pizza	Supreme	Spinach, Red Onions, Pepperoni, Tomatoes, Art...
five_cheese	The Five Cheese Pizza	Veggie	Mozzarella Cheese, Provolone Cheese, Smoked ...

3. Orders:

Order_id(Integer): The unique identifier assigned to each customer transaction. Primary Key, Foreign Key to order_details.csv

Date (Date): The calendar date on which the order was officially placed.

Time (Time): The specific time of day when the order was placed, crucial for analyzing peak demand.

order_id	date	time
1	2015-01-01	11:38:36
2	2015-01-01	11:57:40
3	2015-01-01	12:12:28
4	2015-01-01	12:16:31
5	2015-01-01	12:21:30
6	2015-01-01	12:29:36
7	2015-01-01	12:50:37
8	2015-01-01	12:51:37
9	2015-01-01	12:52:01
10	2015-01-01	13:00:15
11	2015-01-01	13:02:59
12	2015-01-01	13:04:41
13	2015-01-01	13:11:55
14	2015-01-01	13:14:19
15	2015-01-01	13:33:00
16	2015-01-01	13:34:07
17	2015-01-01	13:53:00
18	2015-01-01	13:57:08
19	2015-01-01	13:59:09
20	2015-01-01	14:03:08

4.Order_details:

order_details_id(Integer): The unique identifier for a specific line-item entry within an order (e.g., one entry for a Large Pepperoni, another for a Small Veggie) Primary key.

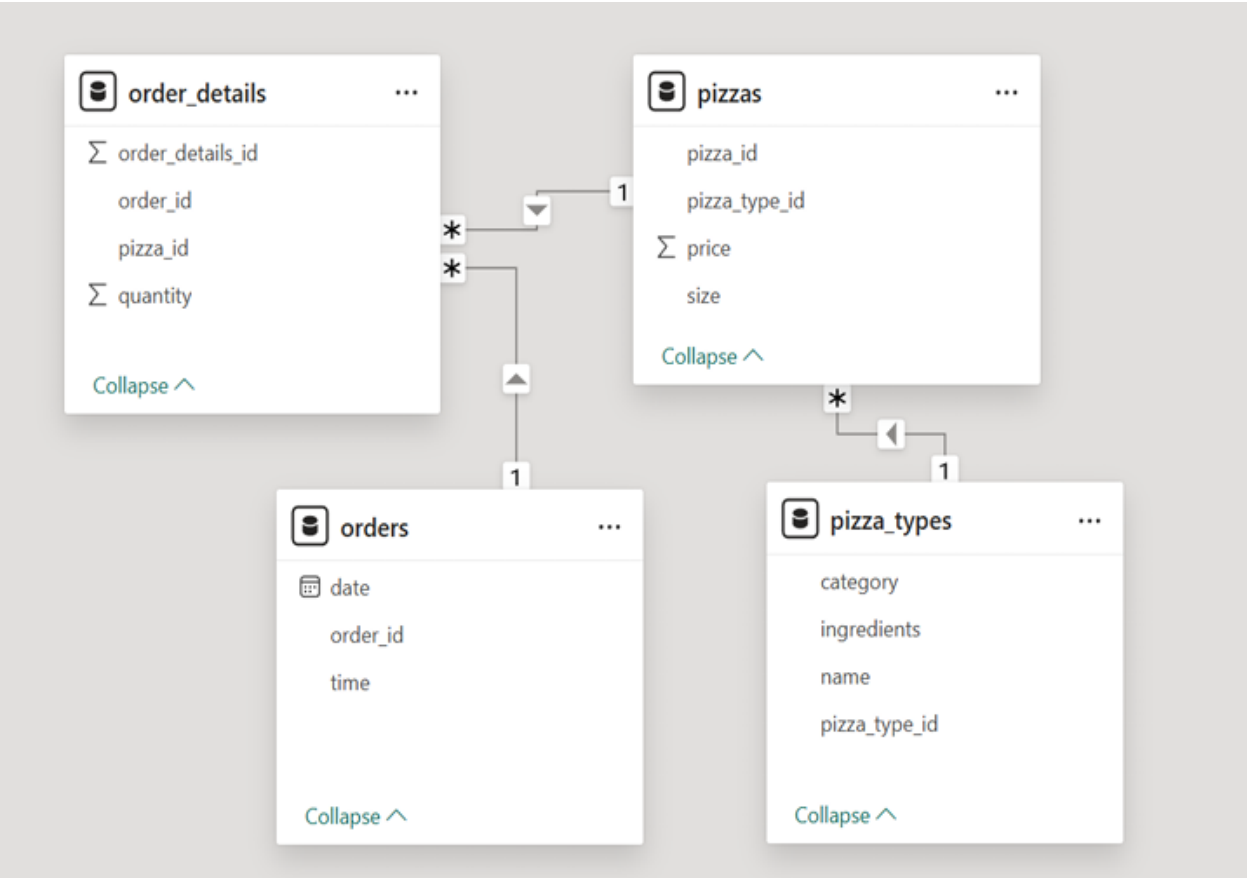
order_id(Integer): An identifier that links this detail back to the high-level order transaction in the orders table. Foreign Key to orders.csv

pizza_id(String): The identifier for the specific pizza variant (type and size) that was ordered, linking to the pizzas table. Foreign Key to pizzas.csv

Quantity (Integer): The number of units of that specific pizza variant ordered within this single line item.

order_details_id	order_id	pizza_id	quantity
1	1	hawaiian_m	1
2	2	classic_dlx_m	1
3	2	five_cheese_l	1
4	2	ital_supr_l	1
5	2	mexicana_m	1
6	2	thai_ckn_l	1
7	3	ital_supr_m	1
8	3	prsc_argla_l	1
9	4	ital_supr_m	1
10	5	ital_supr_m	1
11	6	bbq_ckn_s	1
12	6	the_greek_s	1
13	7	spinach_supr_s	1
14	8	spinach_supr_s	1
15	9	classic_dlx_s	1
16	9	green_garde...	1
17	9	ital_cpdlo_l	1
18	9	ital_supr_l	1
19	9	ital_supr_s	1
20	9	mexicana_s	1
21	9	spicy_ital_l	1
22	9	spin_pesto_l	1
23	9	veggie_veg_s	1
24	10	mexicana_l	1

MODEL OVERVIEW



Case Study Questions

Basic Analysis:

1. Find the total number of orders placed.

Query:

```
use pizzahut;
select count(order_id) as 'Total_Orders'
from orders;
```

Output:

	Total_Orders
▶	21350

Insight:

The Total number of orders placed by customers are 21,350. Indicates a strong customer base and consistent demand across the sales period.

2. Calculate the total revenue from pizza sales.

Query:

```
select
sum(price*quantity) as 'total_revenue'
from order_details as o join pizzas as p
on o.pizza_id = p.pizza_id;
```

Output:

	total_revenue
▶	817860.049999993

Insight:

The Total Revenue Generated is $817,860.0499 Shows healthy sales performance, suggesting effective pricing and steady order volume.

3. Identify the highest-priced pizza.

Query:

```
select
pizza_id,price
from pizzas
order by price desc
limit 1;
```

Output:

	pizza_id	price
▶	the_greek_xxl	35.95

Insight:

The Most Expensive Pizza is "The Greek XXL" with price $35.95. Represents the premium offering on the menu, suitable for luxury marketing or combo deals. Tracking how often this is ordered could reveal high-end customer preferences.

4. Determine the most frequently ordered pizza size.

Query:

```
select
p.size,
count(od.order_details_id) as order_count
from order_details od join pizzas p
on od.pizza_id = p.pizza_id
group by p.size
order by order_count desc
limit 1;
```

Output:

	size	order_count
▶	L	18526

Insight:

The Most Ordered pizza size is Large. It reflects customer behavior — group and family orders dominate.

5. List the top 5 pizzas by order quantity.

Query:

```
SELECT
    pt.name AS pizza_name,
    SUM(od.quantity) AS total_quantity
FROM order_details od JOIN pizzas p
ON od.pizza_id = p.pizza_id
JOIN
    pizza_types pt
    ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY total_quantity DESC
LIMIT 5;
```

Output:

pizza_name	total_quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

Insight:

The most frequently ordered pizzas highlight customer favorites and menu strengths. These items can be promoted through combo deals or featured in marketing campaigns.

Intermediate Analysis:

1.Calculate the total quantity ordered for each pizza category.

Query:

```sql
SELECT
    pt.category,
    SUM(od.quantity) AS total_quantity
FROM order_details od JOIN pizzas p
    ON od.pizza_id = p.pizza_id
JOIN
    pizza_types pt
    ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.category
ORDER BY total_quantity DESC;
```

Output:

category	total_quantity
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

Insight:

The Classic pizza category is the clear bestseller, accounting for 14,888 pizzas sold. This indicates that traditional, simple, and likely lower-priced pizza types are the most popular choice for customers.

The Chicken category has the lowest total quantity sold (11,050). This might suggest these items are specialty, higher-priced, or simply less demanded than the other categories.

2. Analyze the distribution of orders by hour of day.

Query:

```sql
select
HOUR(time) AS order_hour,
    COUNT(order_id) AS total_orders
FROM
    orders
GROUP BY
    order_hour
ORDER BY
    order_hour;
```

Output:

order_hour	total_orders
9	1
10	8
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642
21	1198
22	663
23	28

Insight:

The sales data clearly shows two distinct peak ordering periods centered around mealtimes: the busiest period is lunch (12 PM to 1 PM) and the secondary, sustained peak is dinner (5 PM to 7 PM).

3. Determine the order distribution of pizzas by category.

Query:

```sql
SELECT
pt.category,
COUNT(od.order_details_id) AS total_orders
FROM order_details od JOIN pizzas p
ON od.pizza_id = p.pizza_id
JOIN
pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.category
ORDER BY total_orders DESC;
```

Output:

	category	total_orders
▶	Classic	14579
	Supreme	11777
	Veggie	11449
	Chicken	10815

Insight:

The Classic category is the most frequently ordered, accounting for 14,579 separate pizza items, reinforcing its status as the top-selling segment. The Chicken category is the least frequently ordered by item count (10,815), indicating lower overall customer engagement compared to the other three categories.

4. Calculate the average number of pizzas ordered each day.

Query:

```sql
SELECT
    ROUND(AVG(daily_orders), 2) AS avg_daily_pizzas
FROM (

    SELECT
        o.date,
        SUM(od.quantity) AS daily_orders
    FROM
        orders o
    JOIN
        order_details od
        ON o.order_id = od.order_id
    GROUP BY
        o.date
) AS daily_summary;
```

Output:

	avg_daily_pizzas
▶	138.47

Insight:

The analysis shows that Pizza Hut sells an average of 138.47 pizzas per day. This metric is crucial for daily inventory planning and production scheduling.

5. Identify the top 3 pizzas based on revenue.

Query:

```sql
SELECT
    pt.name AS pizza_name,
    ROUND(SUM(od.quantity * p.price), 2) AS total_revenue
FROM order_details od JOIN pizzas p
    ON od.pizza_id = p.pizza_id
JOIN
    pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY total_revenue DESC
LIMIT 3;
```

Output:

	pizza_name	total_revenue
▶	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5

Insight:

The analysis shows that Pizza Hut sells an average of 138.47 pizzas per day. This metric is crucial for daily inventory planning and production scheduling.

Advanced Analysis:

1. Calculate each pizza type's percentage contribution to total revenue.

Query:

```sql
SELECT
    pt.name AS pizza_name,
    ROUND(SUM(od.quantity * p.price), 2) AS total_revenue,
    ROUND(
        (SUM(od.quantity * p.price) /
         (SELECT SUM(od2.quantity * p2.price)
          FROM order_details od2
          JOIN pizzas p2 ON od2.pizza_id = p2.pizza_id)
        ) * 100, 2
    ) AS revenue_percentage
FROM order_details od JOIN pizzas p
ON od.pizza_id = p.pizza_id
JOIN
pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY revenue_percentage DESC;
```

Output:

pizza_name	total_revenue	revenue_percentage
The Thai Chicken Pizza	43434.25	5.31
The Barbecue Chicken Pizza	42768	5.23
The California Chicken Pizza	41409.5	5.06
The Classic Deluxe Pizza	38180.5	4.67
The Spicy Italian Pizza	34831.25	4.26
The Southwest Chicken Pizza	34705.75	4.24
The Italian Supreme Pizza	33476.75	4.09
The Hawaiian Pizza	32273.25	3.95
The Four Cheese Pizza	32265.7	3.95
The Sicilian Pizza	30940.5	3.78
The Pepperoni Pizza	30161.75	3.69
The Greek Pizza	28454.1	3.48
The Mexicana Pizza	26780.75	3.27
The Five Cheese Pizza	26066.5	3.19

Insight:

The top three pizzas are all Chicken variants—are highly profitable, generating over 15.6% of the total revenue alone. Conversely, the bottom performers like "The Brie Carre Pizza" ($11,588.50) contribute minimally at just 1.42% of total revenue, indicating significant revenue concentration in the specialty segment.

2. Track cumulative revenue growth over time

Query:

```sql
SELECT
    date,
    ROUND(SUM(SUM(od.quantity * p.price))
          OVER (ORDER BY o.date), 2) AS cumulative_revenue
FROM orders o JOIN order_details od
ON o.order_id = od.order_id
JOIN
 pizzas p
ON od.pizza_id = p.pizza_id
GROUP BY date
ORDER BY date;
```

Output:

	date	cumulative_revenue
▶	2015-01-01	2713.85
	2015-01-02	5445.75
	2015-01-03	8108.15
	2015-01-04	9863.6
	2015-01-05	11929.55
	2015-01-06	14358.5
	2015-01-07	16560.7
	2015-01-08	19399.05
	2015-01-09	21526.4
	2015-01-10	23990.35
	2015-01-11	25862.65
	2015-01-12	27781.7
	2015-01-13	29831.3
	2015-01-14	32358.7
	2015-01-15	34343.5
	2015-01-16	36937.65
	2015-01-17	39001.75
	2015-01-18	40978.6
	2015-01-19	43365.75

Insight:

The Revenue Growth is consistent and strong daily. Tracking this daily cumulative revenue is essential for understanding the overall financial momentum and setting long-term growth targets.

3. Determine the top 3 pizzas by revenue within each category.

Query:

```sql
SELECT category,pizza_name,total_revenue
FROM (
    SELECT
        pt.category,
        pt.name AS pizza_name,
        ROUND(SUM(od.quantity * p.price), 2) AS total_revenue,
        RANK() OVER (
            PARTITION BY pt.category
            ORDER BY SUM(od.quantity * p.price) DESC
        ) AS revenue_rank
    FROM order_details od JOIN pizzas p
        ON od.pizza_id = p.pizza_id
    JOIN
        pizza_types pt
        ON p.pizza_type_id = pt.pizza_type_id
    GROUP BY
        pt.category, pt.name
) ranked_pizzas
WHERE revenue_rank <= 3
ORDER BY category, total_revenue DESC;
```

Output:

category	pizza_name	total_revenue
Chicken	The Thai Chicken Pizza	43434.25
Chicken	The Barbecue Chicken Pizza	42768
Chicken	The California Chicken Pizza	41409.5
Classic	The Classic Deluxe Pizza	38180.5
Classic	The Hawaiian Pizza	32273.25
Classic	The Pepperoni Pizza	30161.75
Supreme	The Spicy Italian Pizza	34831.25
Supreme	The Italian Supreme Pizza	33476.75
Supreme	The Sicilian Pizza	30940.5
Veggie	The Four Cheese Pizza	32265.7
Veggie	The Mexicana Pizza	26780.75
Veggie	The Five Cheese Pizza	26066.5

Insight:

Top 3 revenue-generating pizzas within each of the four categories, confirming the high value of the Chicken category, which includes the top overall pizza. This granular revenue ranking is critical for category-specific menu engineering and promotion, ensuring marketing spend is focused on the highest-yield items in every segment.

ACTIONABLE SALES INSIGHTS

- The business is performing well, with over 21,350 total orders placed and a daily sales average of 138.47 pizzas. Use the average daily pizza sale metric (138.47) as the baseline production target and utilize the cumulative revenue trend to set and monitor aggressive, data-informed goals for future growth.

- The Chicken category, while having the lowest order volume, generates significantly higher overall revenue, confirming these items are premium with superior margins. Prioritize marketing and display space for high-revenue pizzas and strategically explore expanding the premium Chicken menu to capitalize on this high-value customer segment.

- Order volume is heavily concentrated around two key meal periods: the absolute peak at Lunch (12 PM – 1 PM) and a sustained peak in the Evening (5 PM – 7 PM). Implement dynamic staffing models to heavily staff the kitchen and delivery teams during the 11 AM to 2 PM and 4 PM to 7 PM windows to minimize customer waiting times and improve throughput.

- The Classic pizza category is the most popular by quantity, and the Large size is the most frequently ordered, indicating a preference for simple, value-driven, and family/group dining options.

- The Veggie and Supreme categories show solid, but not market-leading, order volume, and may be lagging in revenue. To increase the profitability of the Veggie segment, consider introducing new, higher-priced variants (e.g., pizzas with premium cheese or specialty, spicy ingredients) to raise the average ticket size for this popular customer group.

RECOMMENDATIONS

- **Strategic Pricing**: Maintain or slightly increase the price point of these premium items, as their high revenue contribution suggests they can sustain a higher margin despite lower overall volume.

- **Expand Premium Line:** Explore introducing new, high-margin, specialty Chicken items to further capture revenue from customers willing to pay a premium for unique flavors.
- **Aggressive Staffing:** Implement a scheduling model that significantly increases kitchen and delivery staff starting 1 hour before the peak rush periods.
- **Preparation Strategy:** Pre-prep core ingredients required for the best-selling Classic and Large-size pizzas during off-peak hours to ensure rapid assembly during the rush, maintaining high customer satisfaction.
- **Strategic Discounting**: If necessary, use small, targeted promotions (e.g., a "Supreme Monday" deal) to drive incremental volume in the Supreme category, which is the second most popular by quantity, without significantly cutting margins.
- **Evaluate Low Performers:** Review the bottom-performing pizzas (like "The Brie Carre Pizza") and their associated ingredient costs to determine if they should be removed from the menu to reduce complexity, lower ingredient waste, and free up kitchen focus for high-demand items.

CONCLUSION

The comprehensive SQL analysis successfully provided a data-driven blueprint for enhancing Pizza Hut's performance. The study revealed a clear dual strategy prioritizing Classic pizzas to meet high volume demand and aggressively promoting Chicken pizzas to capitalize on their high revenue contribution.

By focusing on menu optimization such as introducing premium variants in the underperforming Veggie segment and by aligning staffing with the lunch and dinner

peak hours, Pizza Hut can tangibly increase profits and improve customer satisfaction. This project underscores the direct, real-world value of data analytics in formulating effective business decisions.